A-A-53998
10 October 1989

COMMERCIAL ITEM DESCRIPTION

DRILL, PIN HOLE, DENTAL

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for MIL-D-36786B, which is cancelled.

This commercial item description covers spiral (twist) drills suitable for drilling pin holes in the dentin of a tooth for the support of the pin retention restorative techniques.

Salient characteristics.

Classification. Drills shall be of the following classes and sizes as specified.

Classes

Class AL. For use with latch angle handpiece
Class FG. For use with friction grip angle handpiece

Sizes. (Diameter, inches)

Size 1.	0.021
Size 2.	0.024
Size 3.	0.027
Size 4.	0.017
Size 5.	0.0135

Material. Drills of one-piece construction shall be fabricated from tungsten-vanadium tool steel. Drills of two-piece construction shall have the head fabricated from tool steel securely fastened to withstand the required drilling forces inside an aluminum alloy shank.

Style, and design. The drills shall be designed for drilling pin holes in the dentin of a tooth for the support of the pin in pin retention restorative techniques. The drills shall be of one-piece construction or two-piece construction. The style, design and dimensions shall be in accordance with the Defense Personnel Support Center (DPSC) Drawings 22160 and 23318. As an alternate, Class AL, size 1, operative length (mm) 2, shall have a groove approximately 1/16 inch from the distal end of the shank. The groove shall measure 0.003 ± 0.001 inch.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:27Z
Check the source to verify that this is the current version before use.

Hardness. The drill head (working end) shall have a Rockwell hardness of 61 to 65C when tested in accordance with ASTM E18.

Runout. The total indicated runout of the drill shall not exceed 0.0030 inch, when tested as specified herein.

Finish. The operative end of the drill shall have a fine machine finish and shall be properly sharpened. The shank and neck shall have a smooth finish.

Identification marking. The shank of each drill shall be color coded as follows:

Color	Head diameter (inches)
Gold	.027
Black	.024
Silver	.021
Red	.017
Pink	.0135

Workmanship. The drill shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Package (PG). One package containing six drills, of one class, and size as specified, constitutes one unit.

Unit package. The drills shall be packaged in a container designed to prevent damage during shipment or storage.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:27Z
Check the source to verify that this is the current version before use.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, and the lot size.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, at inspection level II. The unit of product for examination purposes shall be one drill. Examination for defects shall include but shall not be limited to those listed below:

Microscopic examination. Examination of drill defects not readily perceptible to the naked eye, shall be made using a microscope at 30 power (30X) magnification.

Defects

Style and design not as specified.
Finish not as specified.
Marking not as specified.
Color coding not as specified.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, at inspection level S-1. The unit of product for test purposes shall be one drill, except for chemical analysis, a sufficient quantity of raw material from each lot shall be sampled.

Tests. Tests shall be conducted to determine compliance with commercial item description requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:27Z
Check the source to verify that this is the current version before use.

Chemical analysis of raw material. In the absence of certificates of quality, the chemical composition of the steel used in the manufacture of the drills shall be determined using Method 111 or 112 of FED-STD-151. In case of a dispute, method 111 shall be used for referee analysis.

Dimensions. Inspection of head and shank diameters, shall be accomplished with regularly calibrated tungsten carbide ring gauges, toolmaker's microscope, or other calibrated instruments of equal accuracy. Neck diameters shall be inspected with a calibrated blade-type micrometer caliper or equivalent. Fitting dimensions shall be checked with appropriate gauges with tungsten carbide faces. Overall length and dimensions shall be checked with a standard tungsten carbide faced micrometer caliper or with a toolmaker's microscope or equivalent.

Runout (Eccentricity). The total indicated runout (TIR) of the drill head shall be measured as follows. The drill shall be held in a V block or precision chuck and rotated slowly. A dial indicator sensitive to 0.005 mm (0.0002 inch), equipped with a knife-edge foot, or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, it shall be mounted with the shaft perpendicular to the axis of the drill head and shall be so positioned that the foot contacts the drill head at the point of smallest diameter. The difference between the maximum and minimum dial indicator readings for at least one full rotation of the drill head shall be taken as the TIR.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:27Z
Check the source to verify that this is the current version before use.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6520-00-017-4930	DRILL, PIN HOLE, DENTAL, Latch Angle Handpiece, 0.021 inch diameter by 0.2 operative length
6520-00-149-0565	DRILL, PIN HOLE, DENTAL, Friction Grip Angle Handpiece, 0.027 inch diameter by 0.08 inch operative length
6520-00-149-0571	DRILL, PIN HOLE, DENTAL, Friction Grip Angle Handpiece, 0.021 inch diameter by 0.08 inch operative length

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:27Z
Check the source to verify that this is the current version before use.

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6520-00-420-0510	DRILL, PIN HOLE, DENTAL, Latch Angle Handpiece, 0.027 inch diameter by 0.08 operative length
6520-00-420-0511	DRILL, PIN HOLE, DENTAL, Latch Angle Handpiece, 0.021 inch diameter by 0.08 inch operative length
6520-00-935-1097	DRILL, PIN HOLE, DENTAL, Latch Angle Handpiece, 0.024 inch diameter by 0.2 inch operative length
6520-00-935-1098	DRILL, PIN HOLE, DENTAL, Latch Angle Handpiece, 0.027 inch diameter by 0.2 inch operative length
6520-01-036-2699	DRILL, PIN HOLE, DENTAL, Latch Angle Handpiece, 0.017 inch diameter by 0.06 inch operative length
6520-01-095-3707	DRILL, PIN HOLE, DENTAL, Latch Angle Handpiece, 0.0135 inch diameter by 0.05 inch operative length
6520-01-119-6397	DRILL, PIN HOLE, DENTAL, Latch Angle Handpiece, 0.017 inch diameter by 0.06 inch operative length

Ordering data. Intermediate/exterior package quantities and labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-2266

LOCATION: NCR Disk No. 7

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:27Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Personnel Support Center
Attn: DPSC-BSTE
2800 South 20th Street
Philadelphia, PA 19101

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:27Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53998	DRILL, PIN HOLE, DENTAL

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26109:272
Check the source to verify that this is the current version before use.